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- Student ID: 111403534
- Github Repository: https://github.com/qqqq555/data-science
- Python Version: 3.12.12
- requirements.txt:

```
pandas==2.2.2
numpy==2.0.2
matplotlib==3.10.0
seaborn==0.13.2
scikit-learn==1.6.1
```

1. Formalize Project Topic, Motivations, and Data Sources

- **Project Topic:** Volunteer Capacity vs Service Demand in Taiwan's Long-Term Care (LTC) 2.0 Nutritional Meal Service.
- **Motivations:** As Taiwan rapidly transitions into a super-aged society, the demand for LTC nutritional meal services (Meals-on-Wheels) is growing exponentially. These meals are critical not just for nutrition, but also as daily wellness checks for solitary elders. However, meal delivery heavily relies on the physical capacity of volunteers. This project investigates whether the recruitment of volunteers is keeping pace with the surging service demand across different counties, aiming to identify localized "service bottlenecks" where the volunteer system might face a potential collapse.
- **Data Sources:** 高齡及長期照顧統計專區-長期照顧類 (https://dep.mohw.gov.tw/dos/cp-5223-62358-113.html#_4.%E9%95%B7%E6%9C%9F%E7%85%A7%E9%A1%A7)
    - 4.1.2 長期照顧十年計畫2.0-營養餐飲服務

Data Loading and Cleaning

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import warnings
warnings.filterwarnings('ignore')
```

```
# --- 1. Data Loading and Cleaning (Preserving 110 Year) ---
file_path = '4.2+長期照顧十年計畫2.0-營養餐飲服務.xlsx'

# ------------------------------------------------------------
# A. Load Historical Data (110~)
# ------------------------------------------------------------
df_hist_raw = pd.read_excel(file_path, sheet_name='歷年(110~)', header=None)

# Extract relevant columns: Col 0 (Year), Col 1 (Total Demand), Col 25
hist_data = df_hist_raw.iloc[:, [0, 1, 25]].copy()
hist_data.columns = ['Year', 'Service_Demand', 'Total_Volunteers']

# FILTER: Only keep rows where 'Year' starts with 3 digits followed by
hist_data = hist_data.dropna(subset=['Year'])
hist_data = hist_data[hist_data['Year'].astype(str).str.match(r'^\d{3}年')]

# Extract integer year
hist_data['Year'] = hist_data['Year'].astype(str).str.extract(r'^(\d{3})')[0].ast

# Convert to numeric safely (Government's '…' will become NaN, which is
hist_data['Service_Demand'] = pd.to_numeric(hist_data['Service_Demand'].astype(st
hist_data['Total_Volunteers'] = pd.to_numeric(hist_data['Total_Volunteers'].astyp

# Calculate burden rate (NaNs will remain NaN)
hist_data['Burden_Rate'] = hist_data['Service_Demand'] / hist_data['Total_Volun


# ------------------------------------------------------------
# B. Load County-Level Data (Year 113)
# ------------------------------------------------------------
df_113_raw = pd.read_excel(file_path, sheet_name='113', header=None)

# Extract relevant columns
county_data = df_113_raw.iloc[:, [0, 1, 25]].copy()
county_data.columns = ['County', 'Service_Demand', 'Total_Volunteers']

# Clean 'County' column
county_data = county_data.dropna(subset=['County'])
county_data = county_data[~county_data['County'].astype(str).str.contains('總計|區

# Remove English text
county_data['County'] = county_data['County'].astype(str).str.replace(r'[a-zA-Z\s

# Convert to numeric
county_data['Service_Demand'] = pd.to_numeric(county_data['Service_Demand'].astyp
county_data['Total_Volunteers'] = pd.to_numeric(county_data['Total_Volunteers'].a

# Drop any incomplete rows ONLY in county data (we need clean data for
county_data = county_data.dropna()

# Calculate burden rate
county_data['Burden_Rate'] = county_data['Service_Demand'] / (county_data['Tota

print("Data successfully loaded! (Year 110 volunteer data is intentionally
```

```
display(hist_data)
display(county_data.head())
```

```
Data successfully loaded! (Year 110 volunteer data is intentionally kept as N
```

	Year	Service_Demand	Total_Volunteers	Burden_Rate
6	110	20544	NaN	NaN
7	111	25745	2706.0	9.514043
8	112	24928	2980.0	8.365101
9	113	24019	2887.0	8.319709

	County	Service_Demand	Total_Volunteers	Burden_Rate
7	新北市	561.0	153.0	3.666664
8	臺北市	1653.0	205.0	8.063411
9	桃園市	1153.0	17.0	67.823130
10	臺中市	4150.0	203.0	20.443340
11	臺南市	1566.0	214.0	7.317754

2. Summary statistics

```
# --- Summary Statistics ---
print("--- Historical Trend Statistics (Year 110-113) ---")
display(hist_data.describe())

print("\n--- County-Level Burden Statistics (Year 113) ---")
display(county_data[['Service_Demand', 'Total_Volunteers', 'Burden_Rate']].descril

# Calculate Growth Rates
# Demand growth from 110 to 113
demand_growth = (hist_data['Service_Demand'].iloc[-1] - hist_data['Service_Demar

# Volunteer growth from 111 to 113 (since 110 is NaN)
first_valid_vol = hist_data['Total_Volunteers'].dropna().iloc[0]
vol_growth = (hist_data['Total_Volunteers'].iloc[-1] - first_valid_vol) / firs

print(f"\n[Key Statistical Findings]")
print(f"* Demand Growth (Year 110-113): {demand_growth:.2f}%")
print(f"* Volunteer Growth (Year 111-113): {vol_growth:.2f}%")
print(f"* The national average burden in Year 113 is {county_data['Burden_l
```

```
--- Historical Trend Statistics (Year 110-113) ---
```

	Year	Service_Demand	Total_Volunteers	Burden_Rate
count	4.000000	4.000000	3.000000	3.000000
mean	111.500000	23809.000000	2857.666667	8.732951
std	1.290994	2287.981789	139.335327	0.676826
min	110.000000	20544.000000	2706.000000	8.319709
25%	110.750000	23150.250000	2796.500000	8.342405
50%	111.500000	24473.500000	2887.000000	8.365101
75%	112.250000	25132.250000	2933.500000	8.939572
max	113.000000	25745.000000	2980.000000	9.514043

```
--- County-Level Burden Statistics (Year 113) ---
```

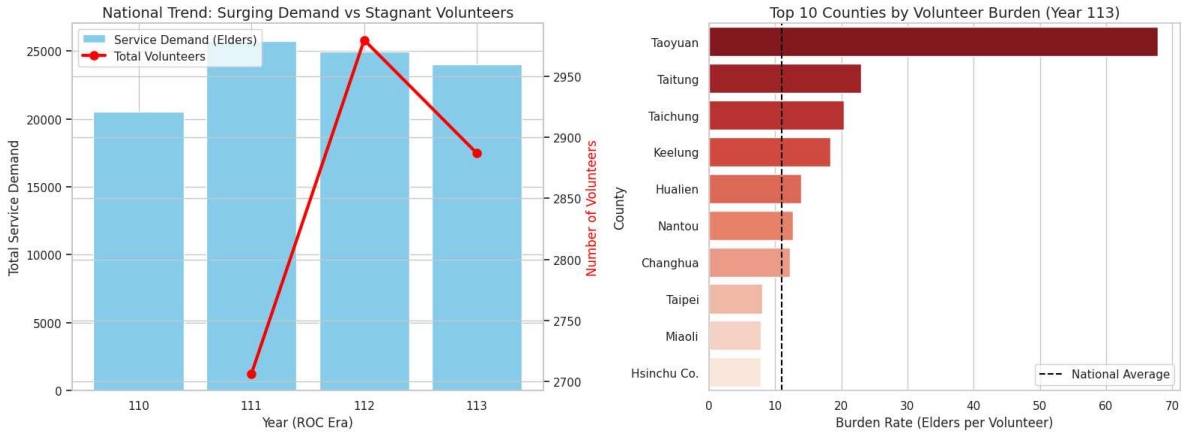
	Service_Demand	Total_Volunteers	Burden_Rate
count	22.000000	22.000000	22.000000
mean	1091.772727	131.227273	10.992287
std	1031.814718	140.811232	14.158560
min	0.000000	0.000000	0.000000
25%	319.750000	45.250000	4.462772
50%	879.500000	94.500000	7.571305
75%	1488.750000	151.000000	12.586999
max	4150.000000	639.000000	67.823130

```
[Key Statistical Findings]
* Demand Growth (Year 110-113): 16.91%
* Volunteer Growth (Year 111-113): 6.69%
* The national average burden in Year 113 is 10.99 elders per volunteer.
```

3. Visualization

```
# --- Visualization ---
# Note: Using standard English fonts to avoid Colab Chinese font renderi
sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Plot 1: Historical Trend (Dual Axis)
ax1 = axes[0]
ax1.bar(hist_data['Year'].astype(str), hist_data['Service_Demand'], color='skyblu
ax1.set_xlabel('Year (ROC Era)', fontsize=12)
ax1.set_ylabel('Total Service Demand', fontsize=12)
ax1.set_title('National Trend: Surging Demand vs Stagnant Volunteers', fonts
```

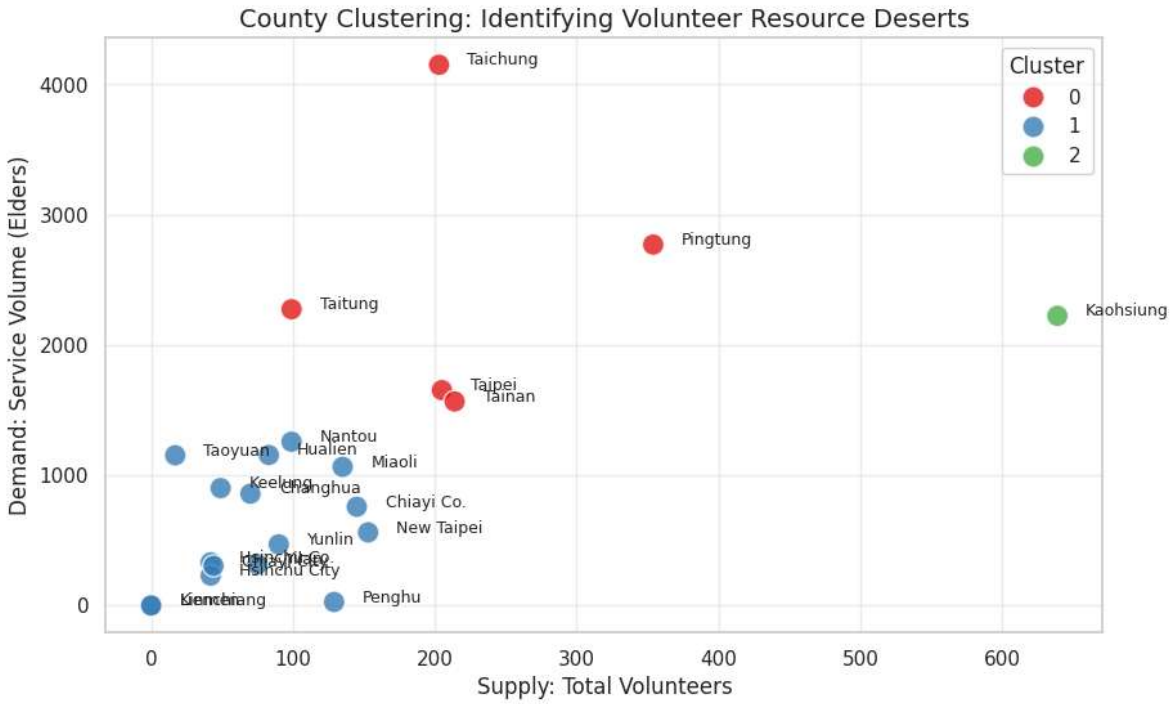
```
ax2 = ax1.twinx()
ax2.plot(hist_data['Year'].astype(str), hist_data['Total_Volunteers'], color='red'
ax2.set_ylabel('Number of Volunteers', color='red', fontsize=12)

# Combine Legends
lines_1, labels_1 = ax1.get_legend_handles_labels()
lines_2, labels_2 = ax2.get_legend_handles_labels()
ax1.legend(lines_1 + lines_2, labels_1 + labels_2, loc='upper left')

# Plot 2: Top 10 Most Burdened Counties
county_sorted = county_data.sort_values('Burden_Rate', ascending=False).head(10)
# Map Chinese county names to English for visualization safety in standa
county_dict = {'新北市': 'New Taipei', '臺北市': 'Taipei', '桃園市': 'Taoyu
county_sorted['County_EN'] = county_sorted['County'].map(county_dict)

sns.barplot(data=county_sorted, x='Burden_Rate', y='County_EN', palette='Reds_r',
axes[1].axvline(county_data['Burden_Rate'].mean(), color='black', linestyle='--',
axes[1].set_title('Top 10 Counties by Volunteer Burden (Year 113)', fontsiz
axes[1].set_xlabel('Burden Rate (Elders per Volunteer)', fontsize=12)
axes[1].set_ylabel('County', fontsize=12)
axes[1].legend()

plt.tight_layout()
plt.show()
```



4. Statistical Modeling: K-Means Clustering

To provide systematic policy recommendations, I apply **K-Means Clustering** based on the `Service_Demand` and `Total_Volunteers`. This machine learning algorithm segments the 22 counties into distinct structural profiles, allowing us to pinpoint "Resource Deserts"—areas that face systemic collapse due to high demand but critically low volunteer numbers.

```
# --- K-Means Clustering ---
# Select features for clustering
features = county_data[['Total_Volunteers', 'Service_Demand']]

# Standardize the data (crucial for distance-based algorithms)
scaler = StandardScaler()
scaled_features = scaler.fit_transform(features)

# Apply K-Means (Segmenting into 3 risk profiles)
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
county_data['Cluster'] = kmeans.fit_predict(scaled_features)

# Map Chinese names to English for the plot
county_data['County_EN'] = county_data['County'].map(county_dict)

# Plotting the clusters
plt.figure(figsize=(10, 6))
sns.scatterplot(data=county_data, x='Total_Volunteers', y='Service_Demand', hue='

# Annotate counties
for i in range(county_data.shape[0]):
        plt.text(county_data['Total_Volunteers'].iloc[i] + 20,
                         county_data['Service_Demand'].iloc[i],
                         county_data['County_EN'].iloc[i], fontsize=9)

plt.title('County Clustering: Identifying Volunteer Resource Deserts', fontsi
plt.xlabel('Supply: Total Volunteers', fontsize=12)
plt.ylabel('Demand: Service Volume (Elders)', fontsize=12)
plt.grid(True, alpha=0.3)
plt.show()

# Display cluster summary
print("--- Average Capacity by Cluster ---")
cluster_summary = county_data.groupby('Cluster')[['Service_Demand', 'Total_Volun
display(cluster_summary)
```




```
--- Average Capacity by Cluster ---
```

Cluster	Service_Demand	Total_Volunteers	Burden_Rate
0	2482.6000	215.0000	13.323807
1	586.4375	73.3125	10.733276
2	2223.0000	639.0000	3.478873

5. Findings and conclusions

- Widening Supply-Demand Gap: From Year 110-113, national demand for LTC nutritional support surged 16.91%, while the volunteer workforce grew only 6.69%. This structural imbalance pushes the national average burden to roughly 11 recipients per volunteer.
- Severe Regional Disparities: Service bottlenecks are highly localized. Taoyuan City faces a critical shortage (67.82 recipients/volunteer), with Taitung, Taichung, and Keelung also showing extreme strain.

- Resource Distribution Profiles (K-Means):
  - Cluster 0 (Resource Deserts): High demand meets low volunteer capacity (e.g., Taipei, Tainan). These areas face the highest risk of systemic burnout (13.32 average burden).
  - Cluster 1 (Volatile/Moderate): Generally lower demand and volunteer numbers, but contains severe individual outliers like Taoyuan.
  - Cluster 2 (Sustainable): Kaohsiung stands alone with balanced scaling, matching high demand with robust volunteer numbers (3.48 average burden).

Actionable Recommendations:

1. Targeted Relief & Innovation: High-burden zones (Cluster 0 & Taoyuan) must pivot from traditional volunteer reliance to alternative delivery models, such as community partnerships or commercial logistics integrations.
2. Strategic Recruitment: Implement a proactive national strategy to overhaul volunteer engagement, ensuring workforce growth aligns with the rapidly aging demographic.
3. Data-Driven Allocation: Continuously monitor these metrics to identify emerging regional bottlenecks before localized systems collapse.